

CSCI 380.61E Web Programming and Interface Design

AWS EC2 + Ubuntu + Nginx + DB + PHP-FPM + React + PM2 Setup Guide

1. Launch AWS EC2 Instance

Recommended instance

- Free-tier
- AMI: **Ubuntu 24.04 LTS** (or 25)

Security Group rules

Port	Purpose
22	SSH
80	HTTP
443	HTTPS
3000 (optional)	Node app dev/test
27017 (optional)	MongoDB remote (NOT recommended—use SSH tunnel)

2. Connect to the EC2 Instance

```
ssh -i yourkey.pem ubuntu@your-public-ip
```

3. Update System

```
sudo apt update && sudo apt upgrade -y
```

4. Install Nginx

```
sudo apt install nginx -y
```

Enable and start:

```
sudo systemctl enable nginx  
sudo systemctl start nginx
```

5. Create Web Root under /srv/www

Ubuntu best practice is under /srv.

```
sudo mkdir -p /srv/www/example.com  
sudo chown -R ubuntu:ubuntu /srv/www/example.com  
sudo chmod -R 755 /srv/www
```

6. Install Database (choose one)

Option A: Install MongoDB

Official repo:

```
sudo apt install -y gnupg curl
curl -fsSL https://pgp.mongodb.com/server-7.0.asc | \
  sudo gpg -o /usr/share/keyrings/mongodb-server-7.0.gpg \
  --dearmor
echo "deb [signed-by=/usr/share/keyrings/mongodb-server-7.0.gpg] \
  https://repo.mongodb.org/apt/ubuntu $(lsb_release -sc)/mongodb-org/7.0 multiverse" \
  | sudo tee /etc/apt/sources.list.d/mongodb-org-7.0.list
```

```
sudo apt update
sudo apt install -y mongodb-org
sudo systemctl enable --now mongod
```

Option B: Install MariaDB (MySQL Alternative)

```
sudo apt install mariadb-server -y
sudo mysql_secure_installation
```

7. Install PHP & PHP-FPM (if needed for backend)

Common PHP packages:

```
sudo apt install php php-fpm php-cli php-mysql php-curl php-zip php-xml php-
mbstring -y
```

Check PHP-FPM:

```
sudo systemctl status php*-fpm
```

8. Install Node.js (LTS)

Use NodeSource:

```
curl -fsSL https://deb.nodesource.com/setup_lts.x | sudo -E bash -
sudo apt install -y nodejs
```

Confirm:

```
node -v  
npm -v
```

9. Install PM2 (Node process manager)

```
sudo npm install -g pm2  
pm2 startup systemd
```

10. Deploy React.js App (Build + Serve with Nginx)

Upload your project:

```
cd /srv/www/example.com  
git clone https://github.com/your/repo.git .
```

Install and build:

```
npm install  
npm run build
```

React build output goes to:

```
/srv/www/example.com/build
```

11. Setup Nginx for React App

Create server block:

```
sudo nano /etc/nginx/sites-available/example.com
```

Paste this:

```
server {  
    listen 80;  
    server_name example.com www.example.com;  
  
    root /srv/www/example.com/build;  
    index index.html;  
  
    location / {  
        try_files $uri /index.html;  
    }  
}
```

```
}

# Reverse-proxy for Node.js API (optional)
location /api/ {
    proxy_pass http://127.0.0.1:3000;
    proxy_http_version 1.1;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection 'upgrade';
    proxy_set_header Host $host;
    proxy_cache_bypass $http_upgrade;
}

# PHP handler (optional)
location ~ \.php$ {
    include snippets/fastcgi-php.conf;
    fastcgi_pass unix:/run/php/php8.3-fpm.sock;
}

access_log /var/log/nginx/example.access.log;
error_log /var/log/nginx/example.error.log;
}
```

Enable your site:

```
sudo ln -s /etc/nginx/sites-available/example.com /etc/nginx/sites-enabled/
sudo nginx -t
sudo systemctl reload nginx
```

12. Deploy Node.js Backend using PM2 (optional)

Example:

```
cd /srv/www/example.com/api
pm2 start server.js --name api-server
pm2 save
```

13. Enable Firewall

```
sudo ufw allow OpenSSH
sudo ufw allow 'Nginx Full'
sudo ufw enable
```

14. Optional: Enable HTTPS (Free SSL via Certbot)

```
sudo snap install core; sudo snap refresh core
sudo snap install --classic certbot
sudo ln -s /snap/bin/certbot /usr/bin/certbot

sudo certbot --nginx -d example.com -d www.example.com
```